

WEEK 5 DBSE \$ DBD

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Execution

1. Created a Spring Boot project with the required dependencies and configured MySQL connectivity.
2. Created the **Student Model, Repository, Service, and Controller** classes.
3. Created the productdb database and connected it with Spring Boot using JPA/Hibernate.
4. Ran the application and verified that the student table was created and student data was inserted.
5. Tested the REST API using `http://localhost:8080/students` and successfully retrieved the student data in JSON format.

```
Spring-Week5-Workspace - Week5-Demo-Spring-MySQL [boot] [devtools]
File Edit Source Refactor Navigate Search Project Run Window Help

Package Explorer
Week5-Demo-Spring-MySQL [boot] [devtools]
  src/main/java
    com.example.demo
      Week5DemoSpringMySQLApplication.java
    com.example.demo.controller
      StudentController.java
    com.example.demo.model
      Student.java
    com.example.demo.repository
      StudentRepository.java
    com.example.demo.service
      StudentService.java
  src/main/resources
    static
    templates
    application.properties
  src/test/java
    com.example.demo
  JRE System Library [JavaSE-21]
  Maven Dependencies
    src
      main
      test
      target
    WCD 0.0.0-1

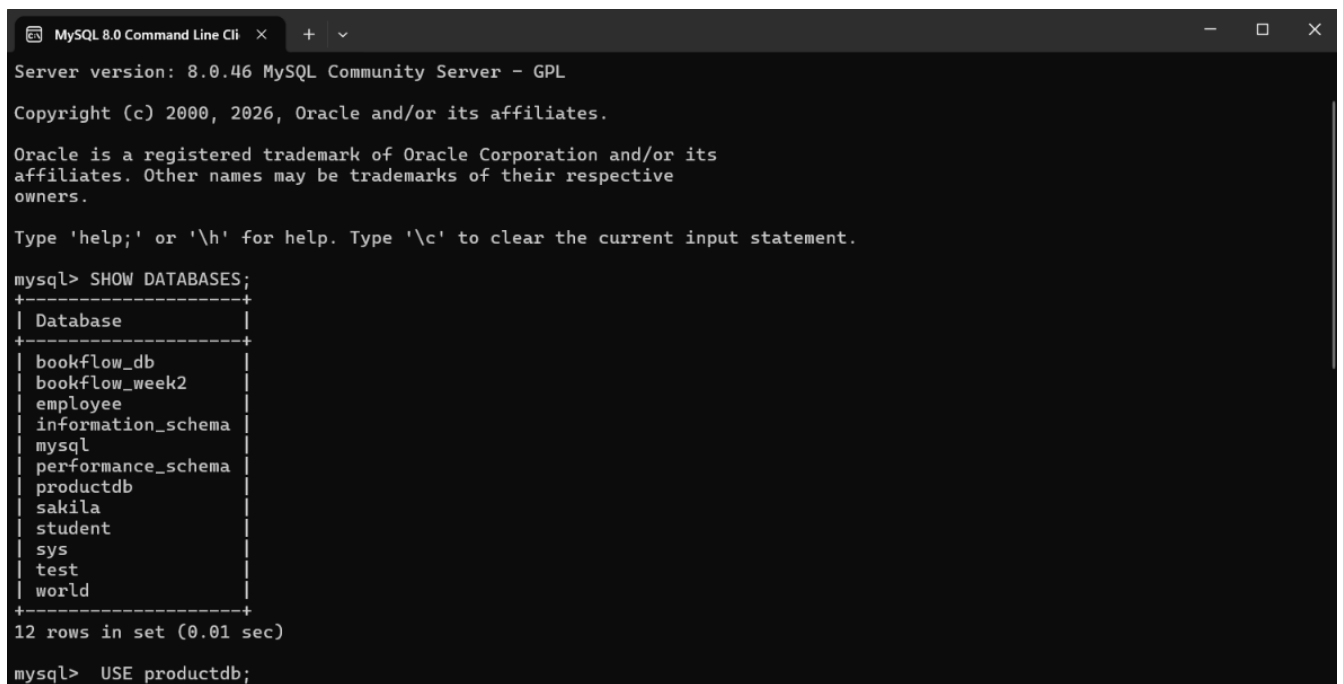
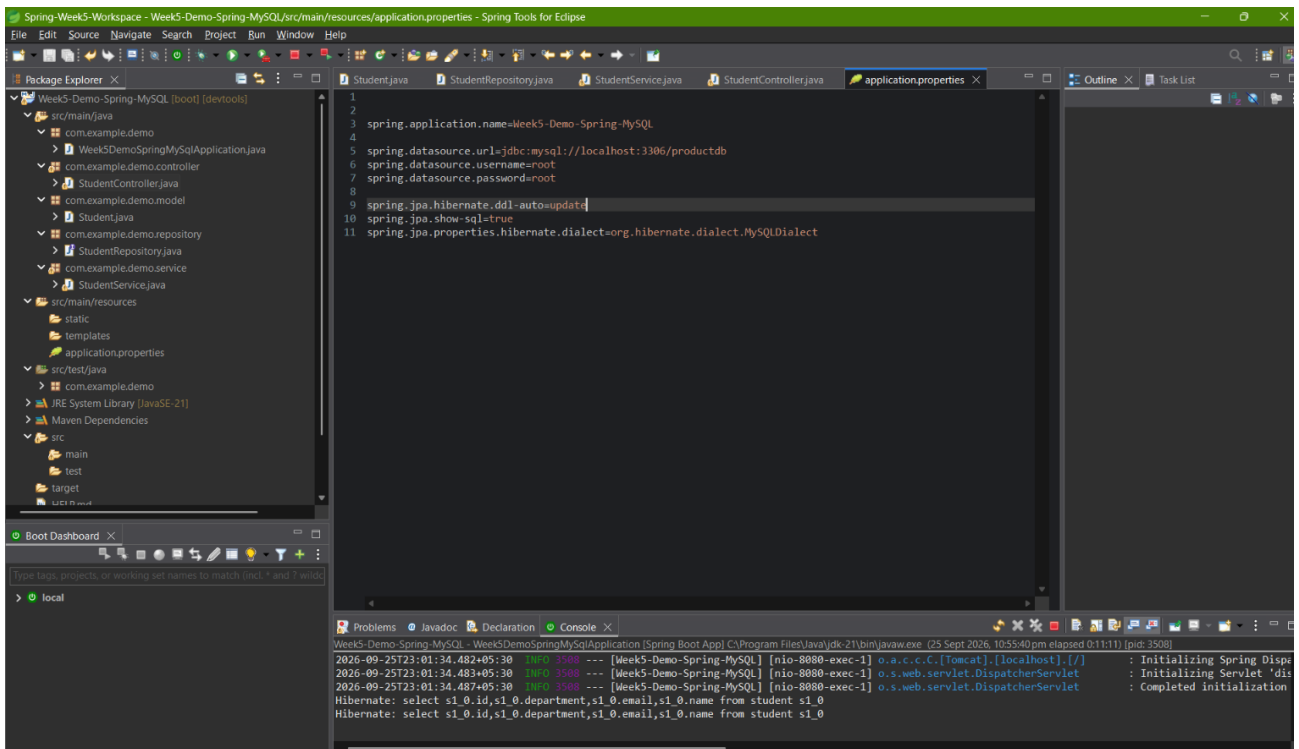
Boot Dashboard
Type tags, projects, or working set names to match (incl. * and ? wildc

> local

StudentController.java
9  import com.example.demo.model.Student;
10 import com.example.demo.service.StudentService;
11
12 @RestController
13 @RequestMapping("/students")
14 public class StudentController {
15
16     @Autowired
17     private StudentService service;
18
19     @GetMapping
20     public List<Student> getAllStudents() {
21         return service.getAllStudents();
22     }
23
24     @GetMapping("/{id}")
25     public ResponseEntity<Student> getStudent(@PathVariable Long id) {
26
27         Student student = service.getStudent(id);
28
29         if (student == null)
30             return ResponseEntity.notFound().build();
31
32         return ResponseEntity.ok(student);
33     }
34
35     @PostMapping
36     public Student addStudent(@RequestBody Student student) {
37         return service.saveStudent(student);
38     }
39
40     @PutMapping("/{id}")
41     public ResponseEntity<Student> updateStudent(
42         @PathVariable Long id,
43         @RequestBody Student student) {
44
45         Student updated = service.updateStudent(id, student);
46     }
47
48 }

Outline
com.example.demo.controller
  StudentController
    service : StudentService
    getAllStudents() : List<Student>
    getStudent(Long) : ResponseEntity<Student>
    addStudent(Long) : Student
    updateStudent(Long, Student) : ResponseEntity<Student>
    deleteStudent(Long) : ResponseEntity<Student>

Console
Week5-Demo-Spring-MySQL - Week5DemoSpringMySQLApplication [Spring Boot App] C:\Program Files\Java\jdk-21\bin\javaw.exe (25 Sept 2026, 10:55:40 pm elapsed 0:10:58) [pid: 3508]
2026-09-25T23:01:34.482+05:30 INFO 3508 --- [Week5-Demo-Spring-MySQL] [nio-8080-exec-1] o.s.c.e.c.[Tomcat].[localhost].[/] : Initializing Spring Dispa
2026-09-25T23:01:34.483+05:30 INFO 3508 --- [Week5-Demo-Spring-MySQL] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dis
2026-09-25T23:01:34.487+05:30 INFO 3508 --- [Week5-Demo-Spring-MySQL] [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization
Hibernate: select s1_0.id,s1_0.department,s1_0.email,s1_0.name from student s1_0
```



```
12 rows in set (0.01 sec)
```

```
mysql> USE productdb;
```

```
Database changed
```

```
mysql> SHOW TABLES;
```

```
+-----+  
| Tables_in_productdb |  
+-----+  
| student              |  
+-----+  
1 row in set (0.01 sec)
```

```
mysql> DESCRIBE student;
```

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
department	varchar(255)	YES		NULL	
email	varchar(255)	YES		NULL	
name	varchar(255)	YES		NULL	

```
4 rows in set (0.01 sec)
```

```
mysql> INSERT INTO student (name, department, email)  
-> VALUES ('Arnav', 'CSE', 'arnav@gmail.com');
```

```
Query OK, 1 row affected (0.03 sec)
```

```
mysql> SELECT * FROM student;
```

```
mysql> DESCRIBE student;
```

Field	Type	Null	Key	Default	Extra
id	bigint	NO	PRI	NULL	auto_increment
department	varchar(255)	YES		NULL	
email	varchar(255)	YES		NULL	
name	varchar(255)	YES		NULL	

```
4 rows in set (0.01 sec)
```

```
mysql> INSERT INTO student (name, department, email)
```

```
-> VALUES ('Arnav', 'CSE', 'arnav@gmail.com');
```

```
Query OK, 1 row affected (0.03 sec)
```

```
mysql> SELECT * FROM student;
```

id	department	email	name
1	CSE	arnav@gmail.com	Arnav

```
1 row in set (0.00 sec)
```

```
mysql> |
```

Created the Student record in SQL and verified

